AI-Guided Database Migration

PostgreSQL ECS to RDS Migration

Cross-region migration with DRS Full + Incremental replication



HUAWEI

Contents

1. **Usage Guidelines** What the AI does, and what stays with you
2. **Your PostgreSQL Scenario** Starting point, goal, and why AI-assisted
3. **What You Need** Prerequisites and the two MCP servers
4. **The Migration Process** Six stages, from check to cleanup
5. **Results and Benefits** What the scenario delivers
6. **Equip Your AI Assistant** Install the skill and start

AI-Assisted Migration: Usage Guidelines

Practical guidance, not a formal delivery commitment

How to Use This Guide

This guide shows what to ask your AI assistant, what knowledge it needs, and how to verify its work to prevent errors. AI accelerates migration planning and execution -- but does not replace expert review or customer decisions.

1. What AI Does

- Discovers your environment, provisions the target, and drives the DRS task.
- Explains every command before you run it
- Does not replace your review or your decisions

2. What stays with you

- Three or four commands on your own server, given to you complete
- Approval at every write operation
- The decision to cut over, and when

3. Your source is read-only

- The AI never connects to your database
- It never modifies your data or configuration
- Your original database keeps running throughout

4. When to Be Cautious

- The database port is briefly open to the internet, restricted to one IP
- For regulated or classified systems, consider private connectivity
- Review every output before acting on it

AI-assisted migration improves efficiency. It does not replace human judgement — verify every step against your own data.

HUAWEI

Your PostgreSQL Migration Scenario

Your starting point

You run PostgreSQL yourself on an ECS. You patch it, back it up, and watch it. It works, but every hour spent on database maintenance is an hour not spent on your product.

Your goal

Move to managed RDS in another region, without losing data and without a long outage. Keep the application running until the moment you decide to switch.

Why AI-assisted

The migration itself is routine. What is not routine is remembering every prerequisite. The AI checks them all first, so failures surface before anything moves.

12 guided steps

6 approval gates

2 MCP servers

0 changes to your source

HUAWEI

What You Need Before Starting

On your side

- PostgreSQL running on an ECS, reachable by public IP
- Logical replication enabled (wal_level = logical)
- A replication user with read access to tables and sequences
- SSH access to that server, for three or four commands
- A target region — the RDS instance is created for you

The AI checks all of this for you in the first step and tells you exactly what to change if something is missing.

Two MCP servers, nothing else

hcloud

Huawei Cloud operations — discovery, DRS task lifecycle, firewall rules. Based on KooCLI.

terraform

Provisioning the target network and the RDS instance.

Both are general-purpose and configured once. No scenario-specific MCP and no browser automation are required.

Example prompt:

"Using this skill and the hcloud and terraform MCPs, migrate my PostgreSQL database from ECS in <source region> to RDS in <target region>. Go step by step and stop at every approval gate."

HUAWEI

Example prompt:

Using this skill and the "hcloud" and "terraform" MCPs, migrate my PostgreSQL database. Go step by step and stop at every approval gate.

SOURCE (already exists — do not modify it, do not connect via SSH):
* Region: xxxx
* Public IP (EIP): xxx.xxx.xxx.xxx
* Port: xxx
* Database name: xxxx
* Replication user: xxxx
* Security group ID: xxxx

TARGET (does not exist yet — create it):
* Region: xxxx

No application to redirect.

Start at Step 0 and wait for my confirmation at each gate. When you need me to run something on the source, give me the complete command and wait.

HUAWEI

The Process: Check, Provision, Connect

1 Check

The AI verifies your database meets the DRS requirements, then records how many rows each table holds. That count becomes the proof later.

Nothing is changed. If something is missing, you get the exact command to fix it.

2 Provision

It asks whether you want a new network or an existing one, proposes a size based on your source, and shows you the Terraform plan before creating anything.

Approval gate: nothing is created until you say so.

3 Connect

It creates the DRS task and opens temporary access between the two databases — firewall rules on both sides, and an access entry you add yourself.

Approval gate: the task is created but no data moves yet.

Verify at this stage: the pre-check must pass with zero blocking items before anything moves. Warnings are listed and explained, and they do not stop the migration.

HUAWEI

The Process: Migrate, Validate, Cut Over

4 Migrate

DRS copies everything, then keeps copying changes as they happen. The full transfer of a small database takes about a minute; replication then runs about a second behind.

Approval gate: this is where data starts moving.

5 Validate

Two checks, both required. Structure is compared automatically. Row counts are compared against the baseline from step 1 — because a perfectly shaped empty database also passes a structure check.

Then a real change is made and watched arriving on the other side.

6 Cut over

You point your application at the new database. Nothing is shut down: your original keeps running, which is exactly what makes this reversible.

Finally the temporary access is removed and the DRS task is stopped, so it stops billing.

Do not skip the row-count check. A structure comparison passes on a target that has every table and no data at all. Only the counts prove the data arrived.

HUAWEI

How You Know It Worked

Two layers of proof, and why both are needed

Structure comparison

- Tables, indexes, constraints and extensions counted on both sides
- Runs automatically as part of the task
- Necessary, but not sufficient on its own

Row-count comparison

- Every table counted and matched against the baseline from step 1
- You run one query and compare the numbers yourself
- This is what actually proves the data moved

Example of a verified result

Table	Source	Target	
demo_customers	6	6	match
demo_orders	6	6	match
demo_order_items	10	10	match
demo_products	5	5	match
Total	**29**	**29**	**match**

Then a live change

One row is added to the source. Thirty seconds later it is counted on the target — without anyone copying it.

That is what confirms replication is live, rather than merely reporting a low delay.

HUAWEI

Why AI-Assisted for This Migration

Failures surface early

Every prerequisite is checked before anything is created. In testing, the three failures that normally cost hours were all caught and prevented up front.

Your source is never touched

The AI does not connect to your database. Commands are handed to you with an explanation. That single rule is what keeps rollback trivial at any stage.

Nothing is left running

The replication task bills while it runs and the firewall stays open until closed. The last step handles both, and tells you what is left on your side.

Measured on two end-to-end runs of this scenario

~1 min	~1 sec	100%	29 / 29
full data transfer	replication delay	pre-check passed	rows verified

Figures from two verified end-to-end executions of this scenario, August 2026. A larger database takes proportionally longer to copy.

HUAWEI

Equip Your AI Assistant

Three steps to run this migration yourself

1 Install the skill

Download the ZIP from the scenario page, or clone the repository. Copy the postgresql-ecs-to-rds-drs folder into your AI tool:

~/.opencode/skills/
~/.claude/skills/
~/.hermes/skills/

2 Connect the two MCPs

hcloud and terraform must both be enabled in your AI tool, and authenticated against your Huawei Cloud account.

Both are general-purpose and are set up once per environment.

3 Start the migration

Send the prompt with your source details — region, public IP, database name, replication user, security group ID — and the target region.

The skill asks for everything else.

Repository

github.com/binrogithub/1-3-Cloud-Adoption-Skills

AI-Assist-Migration / Database-Migration / Relational-Database-Migration

Huawei Proprietary - Restricted Distribution

Thank you.

Bring digital to every person, home and organization for a fully connected, intelligent world.

Copyright©2018 Huawei Technologies Co., Ltd.
All Rights Reserved.

The information in this document may contain predictive statements including, without limitation, statements regarding the future financial and operating results, future product portfolio, new technology, etc. There are a number of factors that could cause actual results and developments to differ materially from those expressed or implied in the predictive statements. Therefore, such information is provided for reference purpose only and constitutes neither an offer nor an acceptance. Huawei may change the information at any time without notice.

